Question #1 of 83

Question ID: 1674698

Which of the following scenarios *most accurately* illustrates the use of liability-driven investing?

Scenario 1:	A commercial bank managing the interest rate exposures of both assets and liabilities on an ongoing basis
Scenario 2:	An asset manager structuring a portfolio of bonds to immunize corporate debt obligations

A) Scenario 2 only. ✓

B) Both Scenario 1 and Scenario 2. ✗

C) Scenario 1 only. ✗

Explanation

Scenario 1 illustrates the use of asset-liability management in managing the interest rate exposures of both assets and liabilities.

In Scenario 2, the asset manager views liabilities as a given and structures assets to manage interest rate risk.

(Module 7.4, LOS 7.g)

Question #2 of 83

Question ID: 1674701

Which of the following portfolios is *most likely* to display the highest convexity?

	Portfolio A	Portfolio B	Portfolio C
Macaulay duration	7.31	7.29	7.30
Cash flow yield	5.64%	5.59%	5.61%
Bond composition (maturity)	2-year bonds and 14-year bonds	9-year bonds	3-year bonds, 8-year bonds, and 12-year bonds

A) Portfolio B. ✗

B) Portfolio C. ✗

C) Portfolio A. ✓

Explanation

For portfolios with the same Macaulay duration and cash flow yield, the portfolio with the greatest dispersion of cash flows will have the highest convexity.

Portfolio A is a barbell portfolio, Portfolio B is a bullet portfolio, and Portfolio C is a laddered portfolio.

Because these three portfolios have essentially the same duration and cash flow yield, barbell Portfolio A will have the highest convexity, and bullet Portfolio B the lowest.

(Module 7.4, LOS 7.i)

Question #3 of 83

Question ID: 1674630

Chris Lipczynski owns a large portfolio that is predominantly invested in domestic equities. He now desires to diversify the portfolio to reduce overall portfolio risk and to generate cash flows that maintain purchasing power over the next 20 years. Lipczynski's current portfolio has a correlation of −0.20 with a domestic fixed-coupon bond portfolio and a correlation of −0.05 with a domestic inflation-linked bond portfolio. The correlation between a domestic fixed-coupon bond portfolio and a domestic inflation-linked bond portfolio is 0.70. Which of the following asset allocations is *most likely* to satisfy Lipczynski's stated desires for his portfolio?

A) 50% domestic equities; 50% fixed-coupon bonds. ✗

B) 50% domestic equities; 30% fixed-coupon bonds; 20% inflation-linked bonds. ✓

C) 80% domestic equities; 20% fixed-coupon bonds. ✗

Explanation

While diversifying into fixed-coupon bonds would offer significant risk-reduction benefits, it would not address Lipczynski's desire to maintain the purchasing power of cash flows over time.

By diversifying into both fixed-coupon bonds and inflation-linked bonds, Lipczynski achieves risk reduction (correlation coefficient between the asset-class pairs are all less than 1), as well as some inflation proofing for the cash flows.

(Module 7.1, LOS 7.a)

Question #4 of 83

Question ID: 1674677

An institutional portfolio manager is managing a fixed-income portfolio with a wide range of bonds and would like to engage in securities lending to enhance returns. Which of the following statements regarding securities lending is correct?

A) Typically, the securities lender must physically deliver the securities to the borrower. ✓

B) The rebate rate is equal to the securities lending rate less the collateral earnings rate. ✗

C) Securities lending arrangements are typically collateralized by cash only. ✗

Explanation

Securities lending often supports short-selling so that in many cases, the security must be physically delivered by the lender to the borrower so that the borrower can immediately cover the short position. The process is achieved through a bilateral repo—the securities borrower specifies the desired security or securities to be received, and the securities lender specifies the amount and types of collateral that will be received back.

Rebate rate = collateral earnings rate – securities lending rate

Security lending arrangements are typically collateralized by cash or high-credit-quality bonds.

(Module 7.3, LOS 7.e)

Question #5 of 83

Question ID: 1674660

A fixed-income analyst gathers the following data on a domestic bond portfolio to forecast the components of the portfolio's total expected return for the next 12 months:

Notional principal	$75 million
Current average bond price	$107.80
Expected average bond price in 12 months if the yield curve remains unchanged	$105.90
Average coupon rate	3.0%
Coupon frequency	Annual

Assuming that there is no reinvestment income, the rolldown return over the 12-month investment horizon is *closest* to:

A) −1.76%. ✓

B) 1.79%.

C) 1.02%.

Explanation

Rolldown return = (expected price at the end of the horizon period − beginning price) / (beginning price) = (105.90 − 107.80) / 107.80 = −1.76%.

(Module 7.2, LOS 7.d)

Question #6 of 83

Question ID: 1674659

Tami Taylor works for a large multinational corporation based in Texas that recently purchased EUR 5 million of Greek bonds from a large shipping conglomerate. Taylor has been called to tomorrow's board meeting to discuss the investment.

Here are selected data on the Greek bond portfolio:

Notional principal of the shipping conglomerate bonds (in millions)	EUR 5.00
Average bond coupon payment (per EUR 100 par)	EUR 5.50
Coupon frequency	Annual
Current average bond price	EUR 94.12
Expected average bond price in one year (assuming unchanged yield curve)	EUR 94.74
Average bond convexity	0.25
Average bond modified duration	5.00
Expected average yield change	+0.50%
Expected yield spread change	+0.20%
Expected USD appreciation versus EUR	0.25%

Assuming no reinvestment income, the expected return of the EUR bond portfolio for the next year is *closest* to:

A) 3.75%.

B) 3.25%.

C) 2.75%.

Explanation

Total expected return = rolling yield + *E*(change in price based on investor's yield view) + *E*(change in price based on investor's yield spread view) + *E*(currency gains or losses)

Rolling yield = yield income + rolldown return

Return Component	Formula
Yield income	Annual coupon payment / current bond price
+ rolldown return	$\frac{\left(\text{bond price}_{\text{end}-\text{of}-\text{horizon period}} - \text{bond price}_{\text{beginning}-\text{of}-\text{horizon period}}\right)}{\text{bond price}_{\text{beginning}-\text{of}-\text{horizon period}}}$
= rolling yield	Yield income + rolldown return
+ E(change in price based on investor's yield view)	[–MD × Δyield] + [½ × convexity × $(\Delta\text{yield})^2$]
– E(change in price based on investor's yield spread view)	[–MD × Δspread] + [½ × convexity × $(\Delta\text{spread})^2$]
+ E(currency gains or losses)	Given
= total expected return	

The portfolio's yield income (current yield) is 5.84%. The portfolio has average coupon payments of EUR 5.50 on EUR 100 notional and currently trades at EUR 94.12. The current yield (yield income) is EUR 5.50 / EUR 94.12 = 5.84%.

Assuming we are at the end of Year 1 and we had an unchanged yield curve, the rolldown return is (EUR 94.74 – EUR 94.12) / EUR 94.12 = 0.66%.

The rolling yield (yield income + rolldown return) is 5.84% + 0.66% = 6.50%.

The bond portfolio has a modified duration of 5.0 and a convexity of 0.25. For a 0.50% increase in yield, the expected change in bond price is: $(–5 \times 0.005) + (0.5 \times 0.25 \times 0.005^2) = –0.025 = –2.50\%$.

Similarly, for a 0.20% increase in spread, the expected change in bond price is: $(–5 \times 0.002) + (0.5 \times 0.25 \times 0.002^2) = –0.01 = –1.00\%$.

Tami expects the USD to appreciate by 0.25% over the next year versus the EUR, resulting in a 0.25% loss to the EUR-denominated portfolio; the EUR has depreciated versus the USD.

The total expected return, combing the components, is:

+ 5.84% yield income

+ 0.66% rolldown return

– 2.50% expected change in price based on Tami's view on yields

– 1.00% expected change in price based on Tami's view on yield spreads

– 0.25% expected depreciation of EUR versus USD

= 2.75%.

(Module 7.2, LOS 7.d)

Question #7 of 83

Question ID: 1674634

A total return strategy focuses on either absolute or relative returns rather than matching returns with liabilities. Which of the following statements *most accurately* captures a total return option for bond investors?

A) Enhanced indexing seeks to minimize tracking error by matching duration, credit risk, and sector exposure with a benchmark. ✗

B) Pure indexing focuses on maximizing active return while minimizing tracking error. ✗

C) Active management permits deviations in duration, credit risk, and sector exposure relative to a benchmark. ✓

Explanation

The option for pure indexing is *not correct* because pure indexing minimizes both active returns and tracking error. There is no active attempt to outperform the benchmark. The option for enhanced indexing is *not correct* because enhanced indexing allows mismatched credit exposure and sector weighting relative to a benchmark, but duration is often still matched. The option for active management is *correct* because active management permits manager discretion in all aspects of bond risk management.

(Module 7.1, LOS 7.a)

Question #8 of 83

Question ID: 1674704

A bond portfolio exhibits the following characteristics:

Characteristic 1:	Bond maturities are spread evenly over the yield curve.
Characteristic 2:	Price risk and cash flow reinvestment risk are balanced in the portfolio.
Characteristic 3:	The portfolio provides good liquidity over time.

The portfolio is *most likely* to be a:

A) barbell portfolio. ✗

B) laddered portfolio. ✓

C) bullet portfolio. ✗

Explanation

These are key characteristics of a laddered portfolio. Compared to bullet or barbell portfolios, a laddered bond portfolio is an investment strategy that offers diversification over the yield curve.

(Module 7.4, LOS 7.i)

Question #9 of 83

Question ID: 1674658

A German manufacturing company has just completed the divestiture of a large subsidiary and wishes to retire some of its medium-term debt obligations held by institutional and individual investors globally. A bond tender offer to repurchase the debt in the open market is *most likely* to be:

A) costly to implement. ✓

B) relatively straightforward because most corporate debt securities are highly liquid. ✗

C) the only alternative for the company to immunize its exposure to the debt obligations. ✗

Explanation

In a bond tender offer, the company would buy back the debt liabilities on the open market, paying a premium above the market price. A bond tender offer tends to be difficult and costly to implement, particularly when the bonds are widely held.

Most corporate bonds tend to be illiquid.

The company could alternatively consider cash flow matching or duration matching to immunize its exposure to the liabilities.

(Module 7.1, LOS 7.c)

Question #10 of 83

Question ID: 1674629

The return volatility exhibited by inflation-linked bonds is *most likely*:

A) lower than the return volatility of equities and fixed-coupon bonds. ✓

B) higher than the return volatility of equities and fixed-coupon bonds. ✗

C) lower than the return volatility of equities, but higher than the return volatility of fixed-coupon bonds. ✗

Explanation

The return volatility of inflation-linked bonds is dependent on the volatility of real interest rates, while the return volatility of equities and fixed-coupon bonds is dependent on nominal interest rates.

The volatility of real interest rates is typically lower than that of nominal interest rates.

(Module 7.1, LOS 7.a)

Question #11 of 83

Question ID: 1674685

Which of the following statements relating to fixed-income portfolio management for a tax-exempt investor is *most accurate*?

Statement 1:	The investor's objective is to maximize risk-adjusted pretax returns.
Statement 2:	All else equal, the investor should prioritize the sale of overvalued bonds over undervalued bonds.

A) Only Statement 1 is accurate. ✗

B) Both Statement 1 and Statement 2 are accurate. ✓

C) Only Statement 2 is accurate. ✗

Explanation

Tax considerations such as tax-loss harvesting are irrelevant for a tax-exempt investor.

(Module 7.3, LOS 7.f)

Question #12 of 83

Question ID: 1674663

Selected data relating to a diversified portfolio of euro-denominated bonds are shown in the following table:

Notional principal	€200 million
Current average bond price	€108.65
Average coupon rate	2.0%
Average effective duration	5.72
Average effective convexity	18

Based on this data and an expected average yield and yield spread change of 42 basis points, the expected change in the portfolio's price is *closest* to:

A) −2.04%.

B) 2.42%.

C) –2.39%.

Explanation

The expected change in portfolio price based on the expected average yield and yield spread change = (−effective duration × Y) + (0.5 × effective convexity × Y^2) = (−5.72 × 0.0042) + (0.5 × 18 × 0.0042^2) = −2.39%.

Effective duration and effective convexity must be used for this calculation if the portfolio contains bonds with embedded options.

(Module 7.2, LOS 7.d)

Question #13 of 83

Question ID: 1674686

When managing a fixed-income portfolio for a taxable investor, which of the following statements is *most accurate*?

Statement 1:	Taxable investors should realize capital losses early and defer the realization of capital gains.
Statement 2:	Taxable investors should evaluate the differences in the taxation of interest income and capital gains when selecting portfolio assets.

A) Only Statement 2 is accurate.

B) Only Statement 1 is accurate.

C) Both Statement 1 and Statement 2 are accurate.

Explanation

Statement 1 refers to tax-loss harvesting.

Statement 2 is an important consideration for a taxable investor (e.g., when there is a differential between the tax rate of interest income and the tax rate of capital gains).

(Module 7.3, LOS 7.f)

Question #14 of 83

Question ID: 1674705

You manage a bond portfolio with a benchmark duration of 8. You are considering the following possible strategies:

- **Ladder.** Replicate the benchmark with an equal distribution of bonds with durations ranging from 1 to 16 (for a combined duration of 8). The ladder option has convexity of 25.6.
- **Barbell.** This option invests 50% in bonds with a duration of 2 and 50% in bonds with a duration of 14 (for a combined duration of 8). The barbell option has convexity of 29.1.
- **Bullet.** Invest 100% in bonds with a duration of 8 and convexity of 21.2.

You are expecting a large, parallel, and upward shift in the yield curve. Which portfolio should you pick?

A) Pick the ladder portfolio. ✗

B) Pick the barbell portfolio. ✓

C) Pick the bullet portfolio. ✗

Explanation

With a large, parallel, and upward shift in the yield curve, bond prices are expected to decline. Convexity will buffer the decline. As such, you should pick the barbell because it has the highest convexity.

(Module 7.4, LOS 7.i)

Question #15 of 83

Question ID: 1674684

Steve Walsh, a portfolio manager at Berkshire Investments, must raise £10 million to meet a client's cash requirements. His client's assets consist of government and corporate bonds in a separately managed account. Interest income and capital gains are taxed at 45% and 20%, respectively, while capital losses in excess of capital gains in the year can be carried forward and used to offset gains in future years. Walsh considers three corporate bonds with the same coupon, maturity, and credit quality for possible liquidation. Selected data on the three bonds are as follows:

	Bond 1	Bond 2	Bond 3
Market value	£10 million	£10 million	£10 million
Unrealized capital gain/loss	−£1.0 million	£0.5 million	£1.0 million

Based on this information, Walsh is *most likely* to sell:

A) Bond 2.

B) Bond 1.

C) Bond 3.

Explanation

Bond 1 should be sold in accordance with a tax-loss harvesting strategy.

A tax-loss harvesting strategy realizes capital losses early and defers the realization of capital gains.

(Module 7.3, LOS 7.f)

Question #16 of 83

Question ID: 1674687

Tara Harris is the lead portfolio manager for the Quantas bond fund and needs to raise $8 million to meet expected withdrawals this quarter. The Quantas bond fund's investors primarily consist of medical professionals who are taxed at the top tax bracket in the United States. Harris is considering selling two bonds this quarter to meet the expected redemptions.

The following are selected data for the two bonds:

	Telecomm Bond	Bank Bond
Current market value	$8 million	$8 million
Capital gain/loss	$200,000	–$400,000
Coupon rate	3.50%	4.00%
Maturity	12 years	12 years
Investment committee analysis	Undervalued	Overvalued

Top U.S. income tax rate	39%
Capital gains tax rate	15%

Based on the information just listed, the optimal tax loss harvesting strategy is:

A) sell $8 million of the bank bond.

B) sell $6 million of the telecomm bond and $2 million of the bank bond.

C) sell $8 million of the telecomm bond.

Explanation

The optimal strategy is to sell $8 million of the bank bond and realize a $400,000 loss for tax considerations. The loss can be used to offset current or future capital gains for tax purposes.

(Module 7.3, LOS 7.f)

Question #17 of 83

Question ID: 1674682

In which of the following investment vehicles does an investment manager's decisions on when to realize capital gains affect the timing of capital gains tax payments by taxable investors?

Investment Vehicle 1:	A mutual fund with pass-through treatment of capital gains.
Investment Vehicle 2:	A separately managed account.

A) Only Investment Vehicle 1.

B) Both Investment Vehicle 1 and Investment Vehicle 2.

C) Only Investment Vehicle 2.

Explanation

When there is pass-through treatment of capital gains in a mutual fund, the investor is taxed when the fund realizes the capital gain.

In a separately managed account, capital gains tax payments are made on realized capital gains when they occur.

(Module 7.3, LOS 7.f)

Question #18 of 83

Question ID: 1674683

Stephanie Joubanian manages a large diversified bond portfolio for a tax-exempt client. Her client now requires $8 million in cash from the portfolio to cover spending needs. Joubanian considers three corporate bonds with the same coupon, maturity, and credit quality for possible liquidation. Selected data on the three bonds are as follows:

	Bond 1	Bond 2	Bond 3
Market value	$8.0 million	$8.0 million	$8.0 million
Unrealized capital gain/loss	−$0.9 million	0	$0.7 million
Investment view	Undervalued	Fairly valued	Overvalued

To meet her client's cash requirements, Joubanian is *most likely* to sell:

A) Bond 1.

B) Bond 3.

C) Bond 2.

Explanation

Bond 3 should be sold because it is overvalued.

Investment decisions for tax-exempt investors are driven by the manager's investment views, as tax considerations are irrelevant.

(Module 7.3, LOS 7.f)

Question #19 of 83

Question ID: 1674688

Which of the following statements about the general taxation of mutual funds in various tax jurisdictions in the world is *most accurate*?

A) Treating realized capital gains that occur in a mutual fund as an increase in the net asset value of the fund shares provides investors with the opportunity to defer the payment of capital gains taxes. ✓

B) Interest income earned within a mutual fund is generally taxed within the fund. ✗

C) The pass-through taxation of gains in mutual funds provides investors with the opportunity to defer the payment of capital gains taxes. ✗

Explanation

By increasing the net asset value of the fund shares each time by the realized capital gains and requiring the payment of taxes by investors only upon the sale of the fund shares, there is a payment deferral opportunity provided.

Interest income earned within a mutual fund is generally taxed in the hands of the investor (not the fund), and it does not matter whether or not the amounts have been distributed to the investor.

The pass-through taxation of capital gains means that any capital gains realized in the fund are "passed through" to the investors in the year that they occur; no deferral opportunity is possible.

(Module 7.3, LOS 7.f)

Question #20 of 83

Question ID: 1674669

Bill McLaughlin is a fixed-income consultant based in the United States. Last year, he forecasted very little changes in the yield curve for the next 14 months with a normal yield curve. McLaughlin concentrated his portfolio on 10-year U.S. Treasury bonds that had a 2.25% coupon, and he purchased his holdings at an average price of 100.38. McLaughlin expected to sell the bonds in one year at 101.92. McLaughlin selected the 10-year maturities because they were the steepest part of the yield curve that he could purchase, because he was constrained to a 10-year maturity from the IPS.

At the start of last year, the expected rolling yield on the portfolio was *closest* to:

A) 3.74%. ✓

B) 2.24%. ✗

C) 1.05%. ✗

Explanation

Rolling yield = yield income + rolldown return.

Yield income = annual coupon payment / current bond price = $2.25 / $100.38 = 2.24%.

Rolldown return = (bond price end – bond price begin) / bond price begin = $101.92 – $100.38 / $100.38 = 1.5%.

Rolling yield = 2.24% + 1.5% = 3.74%.

(Module 7.2, LOS 7.d)

Question #21 of 83

Question ID: 1674651

Which of the following factors is likely to increase the liquidity premium on a bond issue?

Factor 1:	The bond's issuer has a significant number of outstanding bond issues.
Factor 2:	The bond issue will be included in domestic and global bond indices.

A) Only Factor 1. ✓

B) Neither Factor 1 nor Factor 2. ✗

C) Only Factor 2. ✗

Explanation

Factor 1 is likely to lead to a liquidity premium because subsequent bond issues, even if they are of similar maturity and credit quality as previous issues, will now be "off the run," reducing their liquidity. Factor 2 is likely to reflect high liquidity because the index can most likely be traded as an exchange-traded fund (ETF) or pooled investment (mutual fund) and, therefore, reduce bond liquidity premiums.

(Module 7.1, LOS 7.c)

Question #22 of 83

Question ID: 1674624

Which of the following features is provided by fixed-income securities, as an asset class?

A) Their diversification benefits are consistent over time and generalized as an asset class. ✗

B) Fixed-income securities offer an imbedded inflation hedge for investors. ✗

C) The regular cash flows provided by fixed income are vital for large institutions like insurance companies. ✓

Explanation

The option for consistent and generalized benefits is *not correct* because the diversification benefits change in periods of market duress and because the different types of fixed-income securities will have different correlations. The option for the asset-class-level inflation hedge is *not correct* because not all fixed-income securities offer inflation protection. This benefit is reserved for inflation-linked bonds and floating-rate bonds. The option for regular cash flows is *correct* because the stable cash flows offered by fixed-income securities are essential for large institutions, like an insurance company.

(Module 7.1, LOS 7.a)

Question #23 of 83

Question ID: 1674689

Regarding liability-driven investing, which of the following statements is *most correct* relative to Type I liabilities?

A) Modified duration is adequate to model risk exposure. ✓

B) The amount of the payments is known, but the timing of the payments is uncertain. ✗

C) Effective duration is required to model risk exposure. ✗

Explanation

Type I liabilities have known payments and known timing. It is Type II liabilities that include known payments and unknown timing. Simple duration (Macaulay or modified) is adequate for modeling Type I risk exposures. Types II, III, and IV require effective maturity to better capture movements in the yield curve.

(Module 7.4, LOS 7.g)

Question #24 of 83

Question ID: 1674680

Which concept is a good tax management strategy for a bond portfolio manager?

A) Avoid short-term turnover when a bond is trading at a loss. ✗

B) Extend holding periods as long as practical. ✓

C) Focus on high current yielding bonds with short holding periods. ✗

Explanation

The statement about extended holding periods is *correct* because extending holding periods will push capital gains to the longer-term rates, which is the lowest tax rate option. The statement about avoiding short-term turnover is *not correct* because loss harvesting is part of active tax management for a bond portfolio. The statement about high current yielding bonds is *not correct* because income from a bond is taxed at the highest possible tax rate as are short-term gains. This strategy would result in poor tax management.

(Module 7.3, LOS 7.f)

Question #25 of 83

Question ID: 1674666

Jeff Timura is currently the lead portfolio manager for an international fixed-income portfolio. Timura has an upcoming meeting on December 20 with his largest client, Tucker Powell, to discuss the firm's outlook for the upcoming year. At the meeting, Timura tells Powell that he expects the yield curve to continue to remain unchanged for the next 16 months. Assuming the forecast is correct, which of the following statements is *most likely* correct?

A) The current yield (yield income) of an international global portfolio is an incomplete measure of the portfolio's expected return. ✓

B) The best estimate of the bond's return is the portfolio's yield income plus its rolldown return minus its expected credit losses. ✗

C) Because Timura is expecting a stable yield curve, yield income is a good estimate of the bond portfolio's expected return for next year. ✗

Explanation

Yield income (current yield) is an incomplete measure of a bond portfolio's expected return.

Decomposing expected fixed-income returns allows an investor to differentiate among several important return components. At the most general level, expected returns (denoted as E(R) as follows) can be decomposed (approximately) in the following manner:

E(R) ≈ yield income

+ rolldown return

+ E(change in price based on investor's views of yields and yield spreads)

− E(credit losses)

+ E(currency gains or losses)

(Module 7.2, LOS 7.d)

Question #26 of 83

Question ID: 1674653

Which of the following factors is *most likely* to increase the bid-ask spread of a bond issue?

Factor 1:	Greater ease in assessing the bond's credit quality.
Factor 2:	Higher risk associated with the bond.

A) Only Factor 2. ✓

B) Only Factor 1. ✗

C) Both Factor 1 and Factor 2. ✗

Explanation

Greater ease in assessing the bond's quality will typically help to lower the bid-ask spread (e.g., plain vanilla corporate bonds are likely to have lower spreads than callable corporate bonds).

Higher-risk bonds usually have higher bid-ask spreads because of the higher risk that dealers face when holding these bonds in inventory.

(Module 7.1, LOS 7.c)

Question #27 of 83

Question ID: 1674644

Jim Mason is a new associate in operations at a midsized fixed income consulting firm. Mason is trying to learn the benefits of cash flow matching versus duration matching and reaches out to some of his friends on Wall Street for assistance in understanding the key features of liability-based mandates.

Mason's friend, Jeff, tells Mason that the key features of cash flow matching include the following: making no yield curve assumptions, matching the bond portfolio's cash flows to existing liabilities, and frequent rebalancing is not required.

Mason also called his friend, Ted, who tells Mason the benefits of cash flow matching include the following: accommodating multiple yield curve assumptions, frequent rebalancing, and the ability to handle increased complexity, so many advanced traders like this strategy overall.

Which of Mason's friend's explanation of cash flow matching is *most accurate*?

A) Ted's statement.

B) Neither Jeff's nor Ted's statements are correct.

C) Jeff's statement.

Explanation

Jeff's statement is the most accurate. Cash flow matching makes no yield curve assumptions, rebalancing is not required, and bond portfolios are created to match cash flows with liabilities.

(Module 7.1, LOS 7.a)

Question #28 of 83

Question ID: 1674695

A newly hired analyst makes the following statements relating to the classification of liabilities.

Statement 1:	The issuer of floating-rate instruments and putable bonds has Type II liabilities.
Statement 2:	Treasury Inflation-Protected Securities (TIPS) are Type III liabilities from the perspective of the U.S. Treasury.

Which of the analyst's statements are correct?

A) Statement 1 only.

B) Statement 2 only.

C) Both Statement 1 and Statement 2.

Explanation

The issuer of a putable bond has a Type II liability. While the payout amounts are known, there is uncertainty around the timing of payouts (e.g., whether the put option will be exercised by the investor).

Issuers of floating-rate instruments and TIPS have Type III liabilities. While the timing of payouts is known, the payout amounts are uncertain.

(Module 7.4, LOS 7.g)

Question #29 of 83

Question ID: 1674637

The total-return mandate of a bond portfolio specifies that the manager should achieve (1) a return objective of 20 basis points over the benchmark index, (2) active risk of less than 50 basis points, and (3) close matching of the benchmark index's primary risk factors. The mandate *most likely* requires the use of:

A) active management.

B) an enhanced indexing approach.

C) a pure indexing approach.

Explanation

Modest outperformance of the benchmark index, active risk of less than 50 basis points, and close matching of the benchmark index's primary risk factors are typical features of an enhanced indexing approach.

(Module 7.1, LOS 7.a)

Question #30 of 83

Question ID: 1674681

When managing a fixed-income portfolio for a tax-exempt investor, which of the following strategies is *most likely* to be applied?

A) Controlling portfolio turnover.

B) Realizing capital losses to offset capital gains.

C) Extending holding periods to realize long-term, rather than short-term, capital gains.

Explanation

Lower portfolio turnover minimizes transaction costs and boosts portfolio returns.

Strategies such as realizing capital losses to offset capital gains and extending holding periods to defer realizing capital gains are more relevant to taxable investors.

(Module 7.3, LOS 7.f)

Question #31 of 83

Question ID: 1674706

A portfolio has a laddered benchmark and a duration of 8. You are the portfolio manager and you are considering the following possible strategies:

- **Ladder.** Replicate the benchmark with an equal distribution of bonds with durations ranging from 1 to 16 (for a combined duration of 8). The ladder option has convexity of 25.6 and a yield of 4.72%.
- **Barbell.** This option invests 50% in bonds with a duration of 2 and 50% in bonds with a duration of 14 (for a combined duration of 8). The barbell option has convexity of 29.1 and yield of 4.27%.
- **Bullet.** Invest 100% in bonds with a duration of 8, convexity of 21.2, and a yield of 4.92%.

You are expecting a steepening of the yield curve in which the short end of the curve declines by 75 bps, the long end increases by 75 bps, and the intermediate-duration portion remains static. Which portfolio should you pick?

A) Pick the barbell portfolio. ✗

B) Pick the bullet portfolio. ✓

C) Pick the ladder portfolio. ✗

Explanation

The steepening described is effectively a pivot. The increase in the long end of the curve will create a large decline in bond prices. The decrease in the short end will create an increase in prices that does not fully offset the loss on the longer-duration bonds. For this reason, a ladder and a barbell will both experience a net loss. The bullet strategy is estimated to experience no loss because it is concentrated on an intermediate-duration portion of the yield curve that does not move during the pivot described.

(Module 7.4, LOS 7.i)

Question #32 of 83

Question ID: 1674690

Hans Wilsdarf specializes in risk management strategies with Bezel Limited and has been hired to defease $5 million of fixed-income liabilities for a local municipality. The bonds under consideration are putable in May 2021, pay interest semiannually of 4.25%, and mature in 2030. Based on the bond's underlying structure, the amount of cash outlay is known, but the timing of cash outlay is unknown.

These liabilities would be classified as:

A) Type IV. ✗

B) Type II. ✓

C) Type I. ✗

Explanation

Type II liabilities have known amounts, but the timing of those payments is not certain. Examples of Type II liabilities include callable and putable bonds.

(Module 7.4, LOS 7.g)

Question #33 of 83

Question ID: 1674702

Compared to a bullet portfolio with the same duration and cash flow yield, a laddered portfolio is *most likely* to display:

A) lower convexity. ✗

B) lower cash flow reinvestment risk. ✗

C) outperformance in the event of a large, parallel shift in the yield curve. ✓

Explanation

A laddered portfolio, with its dispersion of cash flows, will have a higher convexity than a bullet portfolio with the same duration and cash flow yield.

A laddered portfolio's dispersed cash flows will give rise to higher cash flow reinvestment risk than a bullet portfolio.

A laddered portfolio, with its higher convexity, will outperform a bullet portfolio when there is a large, parallel shift in the yield curve.

(Module 7.4, LOS 7.i)

Question #34 of 83

Question ID: 1674665

Isabelle Macpherson, CFA, reviews the following data related to a well-diversified global bond portfolio for the next year:

Yield income	2.78%
Rolling yield	1.02%
Expected change in portfolio price based on Macpherson's yield change view (falling yield)	0.82%
Expected change in portfolio price based on Macpherson's yield spread change view (falling spread)	0.12%
Expected currency gains	0.35%

The portfolio's total expected one-year return is *closest* to:

A) 0.43%.

B) 2.31%.

C) 5.09%.

Explanation

The rolling yield is the sum of the yield income and the rolldown return.

Total expected return = rolling yield + expected change in portfolio price based on the expected average yield and yield spread change + expected currency gains = 1.02% + 0.82% + 0.12% + 0.35% = 2.31%.

(Module 7.2, LOS 7.d)

Question #35 of 83

Question ID: 1674626

Which of the following statements is *most* descriptive of a horizon-matching bond immunization strategy?

A) The goal of horizon matching is to fund future liabilities with a combination of cash flow matching and duration matching.

B) Horizon matching is a simple option that sacrifices a manager's ability to have flexibility in fixed-income selection.

C) This strategy uses duration matching for short-term liabilities and cash flow matching for longer-term liabilities.

Explanation

The option for duration matching is *not correct* because horizon matching uses cash flow matching in the short term and duration matching in the long term. The option for the goal of horizon matching is *correct* because the goal of horizon matching is to pair the benefits of both cash flow matching and duration matching. The option for lack of flexibility of horizon matching is *not correct* because the duration-matching portion of this strategy is often very complex, but it allows a manager flexibility in asset selection.

(Module 7.1, LOS 7.a)

Question #36 of 83

Question ID: 1674678

Which of the following statements is *most correct* relative to the way that a repurchase (repo) agreement is used to create portfolio leverage?

A) Repos are a long-term funding solution for short sellers. ✗

B) Repos require collateral whose value is higher than the amount being borrowed. ✓

C) These agreements require very specific securities to be used as collateral. ✗

Explanation

The statement about long-term funding solutions is *not correct* because repo agreements are extremely short term in nature. They often involve overnight lending. The statement about the use of very specific securities is *not correct* because collateral for a repo agreement is general in nature. The statement about collateral is *correct* because repos offer some measure of security for the lenders because collateral posted must be greater in value than the amount borrowed. In this way, a riskier transaction could hold rates static relative to a less risky loan but require a larger amount of collateral to be posted.

(Module 7.3, LOS 7.e)

Question #37 of 83

Question ID: 1674697

A portfolio manager adopting liability-driven investing is *most likely* to:

A) view assets as a given and structure liabilities to manage interest rate risk. ✗

B) view liabilities as a given and structure assets to manage interest rate risk. ✓

C) focus on managing the interest rate exposures of both assets and liabilities. ✗

Explanation

Asset-liability management focuses on managing the interest rate exposures of both assets and liabilities.

An asset-driven liabilities approach views assets as a given and structures liabilities to manage interest rate risk.

Liability-driven investing views liabilities as a given and structures assets to manage interest rate risk.

(Module 7.4, LOS 7.g)

Question #38 of 83

Question ID: 1674648

Which of the following statements regarding fixed-income investing and pricing is correct?

A) The presence of prepay features is easily accounted for in matrix pricing. ✗

B) Buy-and-hold investors typically select higher-yield bonds, and active investors typically select lower-yield bonds. ✓

C) The fact that an increasing amount of bond trading is now done on exchanges has reduced the bid-ask spreads, especially for less liquid bonds. ✗

Explanation

Buy-and-hold investors have less need for liquidity as they have no plans to sell the bonds; therefore, they may prefer to select less liquid bonds in exchange for higher yield. In contrast, active investors will prefer more liquid bonds, reasoning their active management strategies will generate additional return and compensate for lower initial yield.

The yield to maturity (YTM) of recent trades of bonds with similar features is used to calculate the inferred market price of similar bonds. The presence of prepay features such as a call option makes finding appropriate YTM information more difficult, making inferred price less accurate.

Most bond trading is still done over the counter (in dealer markets), so it means that less liquid bonds trade at higher bid-ask spreads.

(Module 7.1, LOS 7.c)

Question #39 of 83

Question ID: 1674700

A portfolio manager makes the following statements when explaining the construction of a laddered bond portfolio to an individual investor:

Statement 1:	By selecting bonds with maturities spread out over your investment horizon, there should be no further transactions to consider after the initial purchase of bonds.
Statement 2:	Because there is likely to be a bond that is close to maturity at any particular time, the need to sell an illiquid bond to meet a cash flow need is reduced.

Which of the manager's statements is correct?

A) Statement 2 only. ✓

B) Both Statement 1 and Statement 2. ✗

C) Statement 1 only. ✗

Explanation

A laddered portfolio is likely to require reinvestment of the proceeds from maturing bonds.

The laddered approach provides both liquidity and diversification over time; the diversification of cash flows in a laddered portfolio enhances liquidity management.

(Module 7.4, LOS 7.i)

Question #40 of 83

Question ID: 1674691

An investment firm recently issued a 5-year, 5% coupon noncallable bond. It also issued a 5-year index-linked guaranteed investment certificate (GIC) that will make a guaranteed principal payment in five years plus a return based on the S&P 500. How would you categorize these instruments under their appropriate liability types?

	Bond	GIC	
A)	Type II	Type III	✗
B)	Type I	Type IV	✗
C)	Type I	Type III	✓

Explanation

Type I liabilities have known future amounts and known payout dates. A fixed-coupon, option-free bond is an example of a Type I liability.

Type III liabilities have unknown (uncertain) future amounts but known payout dates. Therefore, a GIC with a payout (even if it has a fixed principal component) that is based on an unknown future index value but on a known payout date is an example of a Type III liability.

(Module 7.4, LOS 7.g)

Question #41 of 83

Question ID: 1674640

In the context of total return mandates, which of the following features is *most likely* indicative of pure indexing?

A) There are slight mismatches between portfolio weights and the benchmark. ✓

B) There is very low active risk. ✗

C) Most primary risk factors are closely matched. ✗

Explanation

Most pure bond indexing seeks to *exactly* match *all* of the risk factors of the index (i.e., duration, credit or quality, sectors, and prepayment risks) while still allowing the manager some leeway on the individual bonds selected. Hence, slight mismatches between portfolio weights and the benchmark may occur.

Pure indexing attempts to replicate the performance of a bond index, so it targets *zero* active risk (and zero active return).

(Module 7.1, LOS 7.a)

Question #42 of 83

Question ID: 1674671

A trainee analyst makes the following statements about the use of leverage in fixed-income portfolio management:

Statement 1:	The degree to which leverage affects portfolio returns is dependent on the amount of borrowed funds relative to the portfolio's equity.
Statement 2:	The use of leverage will always increase overall portfolio returns.

Which of the analyst's statements is correct?

A) Both Statement 1 and Statement 2 are correct. ✗

B) Only Statement 1 is correct. ✓

C) Only Statement 2 is correct. ✗

Explanation

The degree to which leverage affects portfolio returns is dependent on the amount of borrowed funds relative to the portfolio's equity (V_B/V_E) and the amount by which the return on invested funds (r_I) differs from the cost of borrowed funds (r_B).

If $r_I > r_B$, leverage will increase the portfolio's return.

If $r_I < r_B$, leverage will decrease the portfolio's return.

(Module 7.3, LOS 7.e)

Question #43 of 83

Question ID: 1674641

Amanda Smith is a new member of a large pension fund, and her team is focused on fixed-income investing. Amanda has been researching inflation-linked bond volatility and how this volatility compares to the volatility found with conventional bonds and equities. After concluding her analysis, Amanda is *most likely* to find that inflation-linked bonds:

A) have the same volatility as conventional bonds, but less volatility than equities. ✗

B) generally have lower return volatility than conventional bonds and equities. ✓

C) generally are more volatile than conventional bonds, but less volatile than equities. ✗

Explanation

Inflation-linked bonds are linked to real interest rates, not nominal rates. Real rates are generally not as volatile as conventional bonds or equities.

(Module 7.1, LOS 7.a)

Question #44 of 83

Question ID: 1674625

Which of the following statements regarding bonds and inflation is *most accurate*?

A) Floating-rate bonds directly adjust both the coupon and the principal of the reference bond. ✗

B) Fixed-coupon bonds offer no direct element of inflation protection. ✓

C) Inflation-linked bonds directly adjust the coupon and indirectly adjust the principal of the reference bond.

Explanation

The statement regarding fixed-coupon bonds is *accurate* because fixed-coupon bonds offer no protection against the effects of inflation. No element of the bond is adjusted as inflation levels adjust in the economy.

The statement regarding floating-rate bonds is *not accurate* because floating-rate bonds only adjust the coupon, while the principal remains static.

The statement regarding inflation-linked bonds is *not accurate* because inflation-linked bonds directly adjust the principal—which, therefore, indirectly adjusts the coupon payments.

(Module 7.1, LOS 7.a)

Question #45 of 83

Question ID: 1674656

Which of the following are common fixed-income derivatives that are traded on exchanges?

A) Bond futures and options on futures.

B) Interest rate swaps and credit default swaps.

C) Interest rate swaps and bond futures.

Explanation

A common alternative to investing directly in bonds is to use fixed-income derivatives. Bond futures and options on futures are traded on exchanges, while interest rate swaps and credit default swaps trade over the counter (OTC). Since the financial crisis, many OTC securities are starting to have similar features compared to exchange-traded products (e.g., initial margin, daily cash settlement).

(Module 7.1, LOS 7.c)

Question #46 of 83

Question ID: 1674676

The following is selected data on the Omega fixed-income mutual fund:

Current average bond price	$105.00
Average modified duration	8.50
Average annual coupon	$1.50
PV of assets (millions)	$120.00
Value of portfolio's equity (millions)	$62.25
Value of borrowed funds (millions)	$30.26
Borrowing rate	1.75%
Return on invested funds	4.25%

The leveraged portfolio return for the Omega fixed-income mutual fund is *closest* to:

A) 5.47%. ✓

B) 9.43%. ✗

C) 5.75%. ✗

Explanation

Return on the leveraged portfolio =

> Return on invested funds + (value of borrowed funds / value of portfolio's equity) × (return on invested funds – borrowing rate)
>
> = 4.25% + ($30.26 million / $62.25 million) × (4.25% – 1.75%) = 5.47%.

(Module 7.3, LOS 7.e)

Question #47 of 83

Question ID: 1674654

Liquidity can be thought of as the ability to make a large transaction quickly with minimal market price impact. Which of the following is *most correct* relative to bond liquidity concerns?

A) Corporate bond offerings within the same issuer often are highly homogenous. ✗

B) The exchange-traded nature of most fixed-income securities enhances their potential liquidity. ✗

C) On-the-run sovereign securities are the most liquid fixed-income securities. ✓

Explanation

The statement about on-the-run sovereign securities is *correct* because on-the-run sovereign fixed income has the highest level of liquidity relative to other categories of fixed-income securities. The statement about corporate bond offerings is *not correct* because corporate bonds offered by the same issuer can be very divergent. Some may offer call features, while others offer different coupon rates and maturities. The statement about being exchange traded is *not correct* because most fixed-income securities are traded over the counter, which reduces liquidity.

(Module 7.1, LOS 7.c)

Question #48 of 83

Question ID: 1674627

When investors display a "flight to quality" in periods of market stress, their investment actions are *most likely* to increase the correlation between:

A) equities and high-yield bonds. ✓

B) government bonds and equities. ✗

C) government bonds and high-yield bonds. ✗

Explanation

A "flight to quality" typically involves buying safer assets, such as government bonds, and selling riskier assets, such as high-yield bonds and equities.

This is likely to increase the correlation between high-yield bonds and equities, while decreasing the correlation between government bonds and equities, as well as the correlation between government bonds and high-yield bonds.

(Module 7.1, LOS 7.a)

Question #49 of 83

Question ID: 1674667

An institutional portfolio manager is attempting to determine the expected return of a bond portfolio. The bonds in the portfolio are mostly option free. The manager also expects a moderate change in interest rates but is unsure of the direction. Which of the following statements is correct?

A) Convexity is likely to be an important consideration. ✗

B) The computation of yield income is based on actual amounts. ✓

C) The easiest portion of the expected return to compute is the rolldown return. ✗

Explanation

Yield income is calculated as the annual coupon amount divided by the current bond price. Both the numerator and denominator are actual amounts as opposed to projections.

The rolldown return is slightly more complicated to compute because it depends on the curve-fitting technique used.

Given that most of the bonds in the portfolio are option free and the change in yield is only expected to be moderate (as opposed to being very large), the convexity effect will be small and will not be an important consideration.

(Module 7.2, LOS 7.d)

Question #50 of 83

Question ID: 1674655

Which of these statements is *most correct* relative to the effects of liquidity on bond portfolio management?

A) The current yields from previous bond transactions may be used to calculate present bond pricing. ✗

B) Current bond pricing data may be based on estimates from historical data. ✓

C) Active bond managers prefer illiquid bonds. ✗

Explanation

The option for illiquid bond preference is *not correct* because active bond managers actually prefer bonds with strong liquidity so they can be sold when the time comes to harvest a profit. Buy-and-hold bond investors are less concerned with liquidity. The option for current bond pricing data is *correct* because many bonds are thinly traded, which requires estimates to be inferred from historical transactions. The option for current yields is *not correct* because it is YTM that is used to infer bond pricing data.

(Module 7.1, LOS 7.c)

Question #51 of 83

Question ID: 1674649

Pierre Labrosse, a trainee bond analyst, has been asked by his supervisor to analyze the factors affecting bond market liquidity. He makes the following notes after his initial research:

Note 1:	The liquidity of sovereign government bonds of different countries is generally similar.
Note 2:	On-the-run issues are typically more liquid than off-the-run issues of comparable bonds.

Which of Labrosse's notes is *most accurate*?

A) Only Note 1 is accurate. ✗

B) Both Note 1 and Note 2 are accurate. ✗

C) Only Note 2 is accurate. ✓

Explanation

The liquidity of a bond issue is closely related to the type of issuer and its credit quality. Sovereign bonds of different credit quality are likely to exhibit different levels of liquidity.

The liquidity of a bond issue is typically highest immediately after issuance.

(Module 7.1, LOS 7.c)

Question #52 of 83

Question ID: 1674692

A bank has both Type II and Type III liabilities. The bank uses modified duration to model its Type II liabilities and effective duration to model its Type III liabilities. Which of the following statements regarding modeling the bank's liabilities is *most accurate*?

A) The bank adequately models only its Type III liabilities. ✓

B) The bank adequately models both its Type II and Type III liabilities. ✗

C) The bank adequately models only its Type II liabilities. ✗

Explanation

Modified duration is a simple duration formula that is only adequate to model Type I liabilities, which have known future amounts at known payout dates.

The three other liability types (Type II, III, and IV) are more complex and have uncertain payout amounts and/or uncertain payout dates, and they require effective duration for modeling. Effective duration can accommodate various yield curve shapes and shifts in the yield curve, and the bank should be using effective duration to model both its Type II and Type III liabilities. Therefore, the bank is adequately monitoring only its Type III liabilities.

(Module 7.4, LOS 7.g)

Question #53 of 83

Question ID: 1674672

Which of the following statements relating to the use of futures contracts in leveraging fixed-income portfolios is *most accurate*?

Statement 1:	Bond futures contracts enable investors to obtain exposure to a large amount of bonds without having to purchase the underlying bonds.
Statement 2:	The leverage obtained by using a futures contract is calculated as the ratio of the margin amount to the notional value of the futures contract.

A) Only Statement 1 is accurate. ✓

B) Only Statement 2 is accurate. ✗

C) Neither Statement 1 nor Statement 2 is accurate. ✗

Explanation

Futures leverage is calculated as the ratio of the futures exposures (in excess of the margin amount) to the margin amount.

(Module 7.3, LOS 7.e)

Question #54 of 83

Question ID: 1674628

A U.S. investor who wishes to construct a dedicated portfolio of domestic bonds over a long-term horizon is *most likely* to use a combination of:

A) mortgage-backed securities (MBSs) and TIPS. ✗

B) floating-coupon Treasury securities and callable corporate bonds. ✗

C) fixed-coupon Treasury securities and TIPS. ✓

Explanation

A dedicated portfolio is constructed of bonds whose cash flows match the timing and magnitude of future liabilities.

The issuer prepayment option in callable bonds and MBSs makes these securities less suited for inclusion in a dedicated portfolio.

(Module 7.1, LOS 7.a)

Question #55 of 83

Question ID: 1674647

An analyst is concerned about how convexity interplays with their portfolio. After conducting some research, they find that convexity:

A) will increase the price reaction when interest rates unexpectedly increase. ✗

B) is positively correlated with yields. ✗

C) will increase the price reaction when interest rates unexpectedly decrease. ✓

Explanation

Convexity decreases the price reaction when interest rates unexpectedly increase. Conversely, it will increase price reactions when rates unexpectedly decrease. Basically, this measurement increases upside capture and decreases downside capture. Convexity is negatively correlated with yield. This means that a bond with high convexity will generally offer a lower yield, and a bond with low convexity will generally offer a higher yield.

(Module 7.1, LOS 7.b)

Question #56 of 83

Question ID: 1674675

Credit risk is a major source of concern for repo agreements. Protection against borrower default can be mitigated by the safety of the underlying security that is used as collateral and the "haircut" that requires additional capital over the principal borrowed amount.

Which of the following statements regarding the haircut is *most accurate*?

A) The size of the haircut is not related to the underlying asset's volatility. ✗

B) Average credit spreads determine the haircut, not volatility. ✗

C) The size of the haircut increases as the volatility of the underlying asset increases. ✓

Explanation

Haircuts for highly rated government securities are typically around 1%–2% and will be higher for more risky securities. As the underlying security increases in volatility, the haircut percentage generally increases to provide further credit protection against the possibility of default by the borrower.

(Module 7.3, LOS 7.e)

Question #57 of 83

Question ID: 1674674

Forced liquidation of a leveraged fixed-income portfolio may be required when the portfolio's value:

A) falls, decreasing the ratio of borrowed funds relative to the portfolio's equity and increasing the portfolio's leverage. ✗

B) rises, decreasing the ratio of borrowed funds relative to the portfolio's equity and decreasing the portfolio's leverage. ✗

C) falls, increasing the ratio of borrowed funds to the portfolio's equity and increasing the portfolio's leverage. ✓

Explanation

When the leveraged portfolio's value falls, V_E will decrease and V_B/V_E (which measures the use of leverage) will increase.

This may induce lenders to demand repayment of the portfolio's borrowed funds—leading to forced liquidation of the portfolio's assets, potentially at unattractive prices.

(Module 7.3, LOS 7.e)

Question #58 of 83

Question ID: 1674668

A bond investor purchases a 5-year bond paying a 6% annual coupon rate on a semiannual basis with a current bond price of $950. Over the next year, the rolldown return from rolling down the yield curve to the expected yield curve is computed as 1.9%. In addition, the expected change in price for the year based on the investor's yield view is 0.8%. Which of the following amounts is *closest* to the total expected return from this yield curve strategy (ignore any reinvestment interest)?

A) 8.7%. ✗

B) 9.9%. ✗

C) 9.0%.

Explanation

Expected return can be computed as yield income + rolldown return + E (change in price based on yield view).

Yield income = annual coupon payment / current bond price = (30 + 30) / 950 = 6.32%.

Total expected return = 6.32% + 1.9% + 0.8% = 9.02%.

Note that yield income is not simply the 6% annual coupon rate. It is also not the same as the yield to maturity of 7.2% (N = 10, PV = –950, pmt = 30, FV = 1,000; solve for I/Y = 3.6% semiannual or 7.2% annual).

(Module 7.2, LOS 7.d)

Question #59 of 83

Question ID: 1674633

Investors may prefer a potentially more lucrative liability-based bond strategy. Which statement *best* describes a contingent immunization strategy?

A) When the present value of the portfolio's liabilities exceeds the present value of the portfolio's assets, this strategy resorts to active bond management.

B) Contingent immunization is a hybrid that merges active bond management and immunization.

C) When the present value of the portfolio's assets exceeds the present value of the portfolio's liabilities, this strategy resorts to immunization.

Explanation

The statement about contingent immunization being a hybrid is *correct* because contingent immunization allows managers to be active when a positive surplus exists and switch to immunization when a deficit is observed. The statement about when to pick immunization is *not correct* because when the present value of the asset exceeds the present value of the liabilities, the portfolio has a surplus, which is when this strategy calls for active bond management. The statement about when to pick active bond management is *not correct* because when the present value of the liabilities exceeds the present value of the assets, the portfolio has a deficit, which is when this strategy calls for immunization.

(Module 7.1, LOS 7.a)

Question #60 of 83

Question ID: 1674694

Defined benefit pension plan obligations are *most likely* to be classified as:

A) Type II liabilities.

B) Type IV liabilities.

C) Type III liabilities.

Explanation

Type IV liabilities will require uncertain amounts of cash outlay, with uncertain timing of cash outlays. Both the future amounts and payout dates of defined benefit pension plan obligations are uncertain.

(Module 7.4, LOS 7.g)

Question #61 of 83

Question ID: 1674657

Which of the following statements listing the typical liquidity of fixed-income subsectors from highest to lowest is *most accurate*?

A) Corporate bonds with shorter maturity, corporate bonds with longer maturity, sovereigns.

B) Corporate bonds with shorter maturity, sovereigns, corporate bonds with longer maturity.

C) Sovereigns, corporate bonds with shorter maturity, corporate bonds with longer maturity.

Explanation

Bond market liquidity is not constant among fixed-income subsectors. Sovereign government bonds are generally more liquid than corporate bonds, and bonds with longer maturities are generally less liquid than shorter-maturity bonds because many purchasers of long-term bonds intend to hold them until maturity (i.e., they stop trading after a while).

(Module 7.1, LOS 7.c)

Question #62 of 83

Question ID: 1674639

The total return mandate of a bond portfolio specifies a return objective of 75 basis points over the benchmark index. The mandate permits the portfolio's duration to deviate significantly from that of the benchmark index. If the mandate also permits the portfolio to exhibit higher turnover than the benchmark index, the portfolio *most likely* requires the use of:

A) a pure indexing approach.

B) active management.

C) an enhanced indexing approach.

Explanation

Significant outperformance of the benchmark index (more than 50 basis points), large risk factor mismatches, and higher turnover relative to the benchmark are typical features of active bond portfolio management.

(Module 7.1, LOS 7.a)

Question #63 of 83

Question ID: 1674696

An investor holding option-free, fixed-rate Treasury securities is *most likely* to classify these investments as:

A) Type I liabilities.

B) neither Type I assets nor Type I liabilities.

C) Type I assets.

Explanation

The Treasury securities have Type I cash flows and are assets from the investor's perspective. Type I assets, such as traditional fixed-rate bonds with no embedded options, have known amounts and payment dates.

(Module 7.4, LOS 7.g)

Question #64 of 83

Question ID: 1674679

Which the following statements relating to active yield curve strategies is correct?

Statement 1:	To increase portfolio duration using leverage, additional bonds with duration greater than that of the current portfolio should be purchased.
Statement 2:	The most efficient way to add convexity to a well-diversified bond portfolio is to shift the distribution of bonds in the portfolio.

A) Statement 2 only.

B) Neither Statement 1 nor Statement 2.

C) Statement 1 only.

Explanation

When using leverage to increase portfolio duration, leverage alone can perform the duration extension; there is no need to use longer-duration bonds.

Adding convexity by shifting the distribution of bonds in a large, well-diversified portfolio is unlikely to be efficient because of the large number of transactions required and because of the illiquidity of many bonds. The use of derivatives to add convexity (e.g., buying calls and selling puts on bonds) would be more efficient.

(Module 7.3, LOS 7.e)

Question #65 of 83

Question ID: 1674703

Compared to a barbell portfolio with the same duration and cash flow yield, a laddered portfolio is *most likely* to display:

A) lower cash flow reinvestment risk.

B) higher convexity.

C) outperformance in the event of a large, parallel shift in the yield curve.

Explanation

Laddered portfolios have relatively high convexities, yet a laddered portfolio will have a lower convexity than a barbell portfolio with the same duration and cash flow yield.

A laddered portfolio's diversification of cash flows will give rise to lower cash flow reinvestment risk than a barbell portfolio.

A laddered portfolio, with its relatively lower convexity, will underperform a barbell portfolio when there is a large, parallel shift in the yield curve.

(Module 7.4, LOS 7.i)

Question #66 of 83

Question ID: 1674635

An investment analyst states that the following conditions are required when implementing a duration-matching approach to a fixed-income mandate:

Condition 1:	The duration of the portfolio of bond assets and the duration of the liabilities must be the same.
Condition 2:	The present value of the portfolio of bond assets must exceed the present value of the liabilities at current interest rates.

Is the analyst correct with respect to the conditions for duration matching?

A) Yes.

B) No. Only Condition 2 is correct.

C) No. Only Condition 1 is correct.

Explanation

Duration matching also requires the present value of the bond asset portfolio and the present value of the liabilities to be the same at current interest rates.

Condition 2 is required specifically for contingent immunization only.

(Module 7.1, LOS 7.a)

Question #67 of 83

Question ID: 1674661

A fixed-income analyst gathers the following data on a domestic bond portfolio to forecast the components of the portfolio's total expected return for the next 12 months:

Notional principal	$75 million
Current average bond price	$92.50
Expected average bond price in 12 months if the yield curve remains unchanged	$95.25
Average coupon rate	1.5%
Coupon frequency	Annual

Assuming that there is no reinvestment income, the rolling yield over the 12-month investment horizon is *closest* to:

A) 2.97%.

B) 4.59%.

C) –2.89%.

Explanation

Yield income = 1.50 / 92.50 = 1.62%.

Rolldown return = (expected price at the end of the horizon period − beginning price) / (beginning price) = (95.25 − 92.50) / 92.50 = 2.97%.

Rolling yield = 1.62% + 2.97% = 4.59%.

(Module 7.2, LOS 7.d)

Question #68 of 83

Question ID: 1674650

An analyst gathers the following data on three U.S. corporate bonds:

	Bond A	Bond B	Bond C
Issue size	80	100	100
Maturity	10 years	5 years	10 years
Credit rating	A	AA	A
Time since issuance	26 weeks	2 days	12 weeks

Based on this information, the bond with the highest liquidity premium is *most likely* to be:

A) Bond A.

B) Bond B.

C) Bond C.

Explanation

Bond A is most likely to have the lowest liquidity of the three bonds and would therefore require the highest liquidity premium.

Small issue size, longer time to maturity, low credit rating, and longer time since issuance tend to result in lower liquidity.

(Module 7.1, LOS 7.c)

Question #69 of 83

Question ID: 1674636

An analyst makes the following statements about the key features of a pure indexing approach to bond portfolio management:

Statement 1:	The portfolio must hold all the bonds in the same proportions as the benchmark index.
Statement 2:	The risk factors of the portfolio and the benchmark index are matched exactly.

Which of the analyst's statements is correct?

A) Both Statement 1 and Statement 2 are correct. ✗

B) Only Statement 1 is correct. ✗

C) Only Statement 2 is correct. ✓

Explanation

A portfolio manager using a pure indexing approach may not be able to hold all the bonds in the same proportions as the benchmark index because some of the bonds in the index may be illiquid.

(Module 7.1, LOS 7.a)

Question #70 of 83

Question ID: 1674699

Which of the following is a benefit of a bond ladder strategy?

A) Ladder portfolios create natural liquidity, which increases the ability to use less liquid bonds. ✓

B) A laddered strategy focuses on price risk at the expense of higher reinvestment risk. ✗

C) Ladder portfolios have less convexity than bullet portfolios. ✗

Explanation

A ladder portfolio is a form of cash flow matching that creates natural liquidity. If the liquidity is not needed for consumption in the short term, then it can be reinvested for even higher returns in the long run. Because the bonds are short term in nature, illiquid bonds can be used as long as full recovery at maturity is expected. Laddered portfolios have more convexity than bullet portfolios. This is a desirable feature when parallel shifts in the yield curve are anticipated. A laddered portfolio balances price risk and reinvestment risk. Price risk is managed because there is often a bond soon to mature when cash flow is needed. Reinvestment risk is managed because the proceeds of bonds that just matured can be reinvested at short- or long-term maturities, which effectively dollar-cost averages the laddered bond portfolio.

(Module 7.4, LOS 7.i)

Question #71 of 83

Question ID: 1674631

Which of the following bond portfolios preserves the purchasing power of coupon payments to the greatest extent? Assume each portfolio is valued at the same absolute dollar amount.

A) Portfolio C (70% fixed-coupon bonds, 5% floating-coupon bonds, 25% inflation-linked bonds). ✗

B) Portfolio B (65% fixed-coupon bonds, 15% floating-coupon bonds, 20% inflation-linked bonds). ✓

C) Portfolio A (80% fixed-coupon bonds, 10% floating-coupon bonds, 10% inflation-linked bonds). ✗

Explanation

The coupons of floating-coupon bonds are inflation protected, while the principal and coupons of inflation-linked bonds are inflation protected.

The level of inflation protection for coupons is given by the percentage of the portfolio in floating-coupon and inflation-linked bonds.

Level of protection offered by Portfolio A = 10% + 10% = 20%

Level of protection offered by Portfolio B = 15% + 20% = 35%

Level of protection offered by Portfolio C = 5% + 25% = 30%

(Module 7.1, LOS 7.a)

Question #72 of 83

Question ID: 1674646

Bond X and Bond Y have the same duration. If Bond Y has the higher convexity, which of the following statements is *most likely* to be correct in the event of a large yield change?

A) Bond Y will experience a larger price increase when yields fall, while Bond X will experience a larger price decrease when yields rise. ✓

B) Bond X will experience a larger price increase when yields fall, while Bond Y will experience a larger price decrease when yields rise. ✗

C) Both bonds will experience the same price change when yields fall and when yields rise. ✗

Explanation

The price of the higher-convexity bond increases more when yields fall and decreases less when yields rise.

Convexity is generally a desirable property of fixed-income bonds. All else equal (i.e., same duration and yield), a more convex bond loses less if the yield goes up and gains more if the yield goes down, as opposed to a less convex bond.

(Module 7.1, LOS 7.b)

Question #73 of 83

Question ID: 1674693

Liability-driven investing (LDI) is a form of an asset-liability management that is *most appropriately* used to meet the investment objectives of:

A) individual investors only. ✗

B) institutional investors only. ✗

C) both institutional investors and individual investors. ✓

Explanation

LDI can be applied to institutional investors (e.g., defined benefit pension funds) and individual investors (e.g., people saving for retirement).

(Module 7.4, LOS 7.g)

Question #74 of 83

Question ID: 1674645

If a new government bond is issued with a negative coupon rate due to low rates of interest, the duration of the new bond will:

A) be longer than its maturity. ✓

B) not be affected by its negative coupon rate. ✗

C) be shorter than its maturity. ✗

Explanation

If bonds are issued with negative coupon rates, the duration of the bonds will be longer than the maturity.

(Module 7.1, LOS 7.b)

Question #75 of 83

Question ID: 1674643

The bond type that protects against inflation for both the coupon payments and the principal received at maturity is:

A) fixed coupon. ✗

B) inflation linked. ✓

C) floating coupon. ✗

Explanation

Inflation-linked bonds (see the following chart):

	Coupon	Par
Fixed-coupon bonds	Not protected	Not protected
Inflation-linked bonds	Protected	Protected
Floating-coupon bonds	Protected	Not protected

(Module 7.1, LOS 7.a)

Question #76 of 83

Question ID: 1674664

Selected data relating to a diversified portfolio of sterling-denominated bonds are shown in the following table:

Notional principal	£120 million
Current average bond price	£97.25
Average coupon rate	2.0%
Average effective duration	6.38
Average effective convexity	23

Based on this data and an expected average yield and yield spread change of −0.29%, the expected change in the portfolio's price is *closest* to:

A) −1.84%.

B) 2.79%.

C) 1.86%.

Explanation

The expected change in portfolio price based on the expected average yield and yield spread change = (−effective duration × ΔY) + (0.5 × effective convexity × ΔY^2) = [−6.38 × (−0.0029)] + [0.5 × 23 × $(-0.0029)^2$] = +1.86%.

Effective duration and effective convexity must be used for this calculation if the portfolio contains bonds with embedded options.

(Module 7.2, LOS 7.d)

Question #77 of 83

Question ID: 1674673

Nick Johnsen manages a well-diversified portfolio of government and corporate bonds. He is considering the use of interest rate swaps and structured financial instruments to leverage his portfolio's returns. If Johnsen expects interest rates to rise over the next five years, which of the following positions is *most likely* to benefit from his interest rate expectations?

A) Fixed-rate receiver in an interest rate swap.

B) Fixed-rate payer in an interest rate swap.

C) Long position in an inverse floating-rate note.

Explanation

A fixed-rate payer/floating-rate receiver in an interest rate swap would benefit from rising interest rates, as the fixed-rate side of the swap would decrease in value, while the floating-rate payments would increase.

A long position in an inverse floating-rate note would benefit from a decline in interest rates, as would the fixed-rate receiver in an interest rate swap.

(Module 7.3, LOS 7.e)

Question #78 of 83

Question ID: 1674638

Jessica Crawford is analyzing the characteristics of three bond funds for her supervisor at Gamma Investment Advisors. All three funds use a total-return approach and the same index as a benchmark. The following table shows selected data relating to the three bond funds and the benchmark index:

	Fund A	Fund B	Fund C	Index
Sector Allocation				
Government and agency	46.4%	47.5%	41.2%	47.8%
Corporate	24.6%	23.7%	28.3%	23.5%
MBS and other ABS	29.0%	28.8%	30.5%	28.7%
Return and Risk Data				
Duration	5.57 years	5.56 years	6.32 years	5.54 years
Yield	1.69%	1.62%	1.75%	1.64%

Based on this information, the fund that is *most likely* to use active management is:

A) Fund B. ✗

B) Fund C. ✓

C) Fund A. ✗

Explanation

Fund C has the largest deviations of sector allocations and duration from the benchmark index.

Fund B is most likely to use a pure indexing approach, as it has slight mismatches of sector allocations from the benchmark and is duration matched to the benchmark.

Fund A is most likely to use an enhanced indexing approach, as it is duration matched to the benchmark, but with slightly larger deviations of sector allocation from the benchmark compared to Fund B.

(Module 7.1, LOS 7.a)

Question #79 of 83

Question ID: 1674662

Michelle Crozier, a fixed-income strategist, is analyzing the expected one-year return of the bond portfolio shown in the following table:

Notional principal	$100 million
Current average bond price	$102.50
Average coupon rate (paid semiannually)	2.25%
Sector allocation	
Government	45.2%
Agency	10.7%
Corporate (plain vanilla)	39.0%
Corporate (callable)	5.1%

Based on this data, Crozier is *most likely* to forecast the expected change in the portfolio's price based on her yield and yield spread view using:

A) modified duration and convexity. ✗

B) effective duration and effective convexity. ✓

C) modified duration. ✗

Explanation

As the portfolio contains bonds with embedded options, effective duration and effective convexity must be used to forecast the expected change in the portfolio's price based on the investor's yield and spread view.

(Module 7.2, LOS 7.d)

Question #80 of 83

Question ID: 1674642

Amy Leslie currently has 40% of her overall portfolio invested in a diversified fixed-coupon bond portfolio with an average duration of eight years. The remainder of her portfolio is evenly spread across both large- and small-cap stocks that are balanced globally across many sectors and countries, primarily through diversified ETFs. Leslie primarily wants to reduce the overall risk of her portfolio and keep up with inflation until she retires in 16 years.

Leslie is a professional photographer and has a limited understanding of the financial markets. During a recent meeting with her financial advisor, Leslie was informed of some changes that she should consider making to her portfolio. Her advisor identified that the correlation coefficient of Leslie's current bond portfolio with her equity holdings is –.05. Also, the correlation coefficient between her equity holdings and an ETF consisting of inflation-linked bonds is .23. Finally, the approximate correlation coefficient between Leslie's current fixed-income portfolio and the proposed inflation-linked ETF is relatively high at .55.

Her advisor recommends that Leslie should diversify her current fixed-coupon bond holdings by adding inflation-linked bonds to her portfolio.

The following action Leslie would take, considering her primary goal is to reduce her portfolio risk and to keep up with inflation until she retires, is *most likely*:

A) rebalance 25% of her fixed-income holdings into inflation-linked bonds. ✓

B) reduce her current bond holdings by 25% and purchase small-cap emerging market stocks with the proceeds. ✗

C) ignore the recommendation because her portfolio is already properly diversified. ✗

Explanation

Rebalancing 25% of her fixed-income holdings into inflation-linked bonds to an already diversified portfolio of bonds and stocks generally results in superior risk-adjusted real portfolio returns. Leslie can also lower the overall risk of her portfolio by adding inflation-linked bonds to her portfolio. The correlation between nominal fixed-coupon bonds and inflation-linked bonds is .55, which is less than 1.0. Adding the inflation-linked bonds helps to meet Leslie's requirement to keep up with inflation.

Increasing the percentage of the portfolio in emerging markets stocks will not reduce the overall portfolio risk as Leslie has requested.

Based on Leslie's primary goal, she should follow her advisor's advice and add inflation-linked bonds to her portfolio.

(Module 7.1, LOS 7.a)

Question #81 of 83

Question ID: 1674652

A disadvantage of matrix pricing for bonds is that it:

A) may be difficult to identify comparable bonds to the bond being priced. ✓

B) cannot be used for illiquid bonds. ✗

C) does not require sophisticated modeling of term structure and credit spreads. ✗

Explanation

Matrix pricing is commonly used to price illiquid bonds.

An advantage of matrix pricing is that it does not require sophisticated modeling of term structure and credit spreads.

(Module 7.1, LOS 7.c)

Question #82 of 83

Question ID: 1674632

Safeera Khan owns a large portfolio that is invested in domestic equities. Khan's portfolio has a correlation of −0.20 with a domestic fixed-coupon bond portfolio and a correlation of −0.10 with a domestic floating-coupon bond portfolio. Khan is *most likely* to diversify her portfolio into domestic floating-coupon bonds to:

A) increase the level of inflation protection provided by the portfolio. ✓

B) maximize the risk-reduction benefits from diversification. ✗

C) fund future liabilities more accurately. ✗

Explanation

The coupons of floating-coupon bonds are inflation protected.

Diversifying into fixed-coupon bonds would provide better diversification benefits (because Khan's portfolio has a more negative correlation with the fixed-coupon bond portfolio) and better funding of future liabilities (because coupon payments are more predictable).

(Module 7.1, LOS 7.a)

Question #83 of 83

Question ID: 1674670

Selected data relating to a fixed-income portfolio are shown in the following table:

Amount of portfolio's equity	£50 million
Amount of borrowed funds	£100 million
Return on invested assets	4.8%
Rate paid on borrowed funds	2.2%

The levered portfolio return is *closest* to:

A) 10.0%.

B) 6.1%.

C) 5.7%.

Explanation

Levered portfolio return = $r_I + [(V_B/V_E) \times (r_I - r_B)] = 4.8\% + [(100/50) \times (4.8\% - 2.2\%)] = 10.0\%$

(Module 7.3, LOS 7.e)